BOPUURU NAVEEN KUMAR
TEST ENGINEER-2 (Storage)
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Summary:

QA Test Engineer with around 4 years of experience in enterprise storage validation, server validation, and test automation. Skilled in validating HDDs, SSDs, NVMe, RAID, Fibre Channel (FC), iSCSI, storage arrays, and enterprise storage platforms. Strong expertise in Python and Pytest automation, Linux debugging, FIO performance testing, firmware validation, fault injection, recovery testing, and log analysis. Experienced in validating BIOS, BMC, CPLD, storage controllers, HBAs, USB, and PCIe, along with protocol debugging, trace analysis, telemetry, JTAG, and firmware debug-log analysis across hardware and firmware configurations. Proficient in developing reusable automation frameworks, integrating regression testing with Jenkins and Git, and leveraging AI-assisted tools such as ChatGPT/OpenAI, GitHub Copilot, and Claude to improve automation, troubleshooting, and validation efficiency.

Professional Experience:

Company Name – **ORNEXT Technology Solutions**

- Present Role: **Test Engineer-2**
- **Client: Hitachi Vantara | April 2022- Present|**

projects:

Feature Validation of Hitachi VSP One Block NVMe Storage System
Client: Hitachi Vantara (Sep 2023 – present)
Project Description:

Worked on feature validation of Hitachi VSP One Block, an NVMe-based high-end all-flash enterprise storage array designed for ultra-high performance and low latency workloads. Performed functional testing of storage features including storage pool creation, RAID level configuration, LUN provisioning, snapshots, and clone operations. Performed failover and multipathing tests to verify high availability and data path redundancy.

Roles & Responsibilities

- Validated NVMe-based storage array features, including NVMe drive discovery, provisioning, namespace management, hot-plug, failover, RAID/data protection, multipathing, health monitoring, performance, firmware upgrades, and fault-recovery scenarios across hardware and firmware configurations.
- Preparing Test bed as per the configuration sheet.
- Installed Operating systems and updated drivers and firmware for HBAs, and RAID controllers to ensure compatibility with Hitachi Vantara storage.
- Validated NVMe storage functionality and health monitoring using NVMe CLI across multiple test environments.
- Performed RAID, storage pool, volume/LUN, hot-spare, degraded-mode, rebuild, recovery, compatibility, firmware, and fault-injection validation.
- Configured RAID levels such as RAID-1, RAID-5, RAID-6, RAID-10.
- Successfully validated failover and multipathing scenarios to ensure high availability and uninterrupted data access.
- Validated volume operations such as create, resize, snapshot and delete.
- Validated FC/iSCSI connectivity, multipathing, path failover, controller failover/failback, and continuous I/O during reboot and power-cycle scenarios.
- Measured IOPS, latency and throughput Preparing using FIO, validating performance against expected benchmarks under mixed workload conditions.

- Automated test cases using pytest to improve testing efficiency and used Python to create test data and automate repeated tests.
- Reduced manual testing effort and improved regression efficiency by automating storage validation test cases using Python and Pytest.
- Used VMware ESXi to create and manage VMs, snapshots, clones, and templates.
- Raising defects through Defect Tracking Tool (JIRA).
- Leveraged ChatGPT/OpenAI and GitHub Copilot for Python automation development, Pytest test generation, code optimization, troubleshooting, and AI-assisted log analysis.

Platform Validation of Hitachi VSP E590H Storage System
Client: Hitachi Vantara (Apr 2022 – Aug 2023)
Project Description

The Hitachi Virtual Storage Platform (VSP) E590H is a midrange enterprise storage family designed to deliver high performance, reliability and efficiency at an affordable cost. Performed Functional testing on Platform-level components including BIOS, BMC and CPLD. Validating key firmware features such as remote system management, sensor monitoring, and system event logging, boot configurations and hardware initialization. Performing firmware upgrade process and ensuring proper integration and stable operation across different hardware configurations to maintain system reliability and performance.

Roles & Responsibilities

- Validated enterprise storage components including controllers, BIOS, BMC, CPLD, SSD/HDD/NVMe, HBAs, PCIe, networking, power supplies, and storage enclosures across hardware and firmware configurations.
- Validated BIOS/BMC functionality including boot, device enumeration, sensors, event logs, remote console, power control, firmware updates, reset, and recovery.
- Executed platform fault and recovery scenarios covering controller/BMC/BIOS resets, reboots, power cycles, firmware failures, and device recovery.
- Validated PCIe devices and interfaces by verifying device enumeration, link status, connectivity, and system compatibility across different hardware configurations.
- Developed and maintained Python/Pytest automation frameworks with reusable fixtures, parametrization, assertions, utilities, and modular test components for scalable server validation.
- Updating firmware for storage array platform components like BMC, BIOS and CPLD.
- Implemented Python automation using os, sys, subprocess, json, datetime, hashlib, and Paramiko for remote execution, process management, log parsing, and test orchestration.
- Worked on multiple platforms including VMware ESXi, Windows server, and Linux environments.
- Integrated automation with Git/Jenkins for scheduled regression execution and leveraged ChatGPT/OpenAI and GitHub Copilot for test generation, code optimization, debugging, and AI-assisted log analysis.

Professional Summary:

- Proficient in Python with strong knowledge of OOP concepts, data structures, file handling, exception handling, regular expressions, and scripting.
- Experienced in developing and maintaining Python/Pytest automation frameworks for storage validation, regression testing, log collection, and result analysis.
- Proficient in Python modules including os, sys, subprocess, argparse, hashlib, logging, datetime, json, re, and Paramiko for automation, remote execution, and log analysis.
- Hands-on experience in validating enterprise storage platforms, focusing on drive qualification, compatibility, firmware, reliability, performance, and fault recovery.

- Strong understanding of Functional, Integration, System, Regression, Performance, Stress, and Re-testing methodologies with knowledge of SDLC, STLC, Agile, and Defect Life Cycle.
- Experienced in designing, executing, and automating test cases for drive validation, RAID validation, firmware validation, and storage feature testing.
- Hands on experience in storage interfaces and protocol validation including USB, PCIe, Teledyne LeCroy Analyzer, PCAP, protocol debugging, JTAG, telemetry, and firmware debug-log analysis.
- Hand on experience in SSD validation, including FTL behavior, wear leveling, garbage collection, firmware debugging, and analytical troubleshooting of storage-related issues.
- Strong knowledge of enterprise storage technologies including HDD, SSD, NVMe, Storage Arrays, JBOD, DAS, SAN, and NAS.
- Familiar with storage protocols including SATA, SAS, SCSI, NVMe, iSCSI, and Fibre Channel (FC).
- Skilled in storage performance analysis using FIO, evaluating IOPS, throughput, latency, bandwidth, and queue-depth metrics.
- Experienced in storage diagnostics using smartctl, nvme-cli, storcli, lsblk, lspci, dmesg, and Linux system logs for troubleshooting and root-cause analysis.
- Good understanding of SSD technologies including wear leveling, TRIM, garbage collection, write amplification, over-provisioning, endurance, and SMART health monitoring.
- Strong knowledge of RAID technologies including RAID levels, Hot Spare, Online RAID Level Migration (ORLM), Capacity Expansion, Consistency Check, Rebuild, and Hot Swapping.
- Proficient with Linux, Shell Scripting, Git, Jenkins, JIRA, PyCharm, PuTTY, VMware, and x86 server firmware components including BIOS, BMC, and CPLD.
- Leveraged GitHub Copilot and Claude to accelerate automation development, log analysis, and debugging.

Academic Credentials:

- **Bachelor of Business Administration (BBA) |** REVA UNIVERSITY **2016 – 2019**

Technical Skills:

Programming & Automation: Python, Pytest, Shell Scripting, OOP, Data Structures, File Handling, Exception Handling, Regular Expressions, Test Automation, Automation Framework Development

Python Libraries & Modules: os, sys, subprocess, argparse, json, datetime, hashlib, logging, re, Paramiko

Storage Technologies: HDD, SSD, NVMe, SATA, SAS, SCSI, RAID, Storage Arrays, Storage Pools, Volumes/LUNs, JBOD, DAS, SAN, NAS, Storage Enclosures, Hot Spare, Hot Swapping

Storage Protocols: Fibre Channel (FC), iSCSI, SATA, SAS, SCSI, NVMe, Multipathing, Path Failover/Failback, Controller Failover/Failback

RAID Technologies: RAID Levels, RAID Configuration, RAID Rebuild, Degraded Mode, Hot Spare, Online RAID Level Migration (ORLM), Online Capacity Expansion, Consistency Check, Hot Swapping, Fault Injection & Recovery

Storage Performance & Testing: FIO, IOmeter, Performance Testing, Stress Testing, Endurance Testing, Reliability Testing, IOPS, Throughput, Latency, Bandwidth, Block Size, Queue Depth, Workload Analysis, Drive Qualification, Compatibility Testing, Firmware Validation

Storage Diagnostics & Troubleshooting: smartctl, nvme-cli, storcli, mdadm, lsblk, lspci, dmesg, Linux System Logs, Log Analysis, Error Analysis, Root Cause Analysis (RCA), Failure Analysis, Performance Analysis

Server & Firmware Validation: x86 Servers, BIOS, BMC, CPLD, Firmware Validation, Firmware Upgrade/Downgrade, Boot Validation, Device Enumeration, Configuration Persistence, System Health Sensors, Event Logs, Remote Console, Power Control, Reset & Recovery, Power Cycle, Reboot, Firmware Failure Recovery

Storage Platform Validation: Enterprise Storage Platforms, Hardware Validation, Firmware Validation, Software Validation, Feature Validation, Functional Testing, Integration Testing, System Testing, Regression Testing, Compatibility Testing, Performance Testing, Stress Testing, Endurance Testing, Fault Injection, Recovery Testing, Drive Qualification

Testing Methodologies: Manual Testing, Automation Testing, Functional Testing, Integration Testing, System Testing, Regression Testing, Re-testing, Performance Testing, Stress Testing, Endurance Testing, Compatibility Testing, SDLC, STLC, Agile, Defect/Bug Life Cycle, Test Case Design, Test Case Execution

CI/CD & Version Control: Jenkins, Git, CI/CD Automation, Scheduled Regression, Automated Test Execution, Result Collection, Failure Reporting

AI & Automation Tools: OpenAI/ChatGPT, GitHub Copilot, Claude, AI-Assisted Test Generation, Code Optimization, Debugging, Log Analysis, Troubleshooting

Operating Systems & Virtualization: Linux, RHEL 9.4, Windows Server 2022/2025, VMware ESXi 8.0/9.0, VMware, Virtual Machines, Snapshots, Cloning, High Availability (HA), Fault Tolerance (FT), Distributed Resource Scheduler (DRS)

Hardware & Server Components: Storage Controllers, HBAs, PCIe, Network Interfaces, Power Supplies, Storage Enclosures, BIOS, BMC, CPLD

Tools: JIRA, PyCharm, PuTTY, Git, Jenkins, FIO, IOmeter, mdadm, storcli, smartctl, nvme-cli, IPMI, Paramiko, VMware

Declaration:

I do hereby declare that the above information is true to the best of my knowledge.